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A note on forecast reconciliation

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ABSTRACT

Forecast reconciliation is a highly effective methodology for improving the predictive accuracy of multiple time series that are subject to linear constraints. This note provides an alternative derivation of the optimal reconciled predictor, based on linear projection arguments, under a set of minimal assumptions on the nature of the reconciliation error. Our result clarifies the relationship between the latter and the preliminary prediction error and encompasses well-known partial reconciliation problems, such as optimal linear disaggregation.

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1. Introduction

Forecast reconciliation applies to linearly constrained time series that are subject to cross-sectional, temporal, and cross-temporal aggregation constraints. It has been a very active area of research with significant influence on the applied forecasting literature. Reconciliation aims to improve the forecast accuracy of the components of a multivariate time series by ensuring coherence with the aggregation constraints. The reconciliation formulae were derived by Hyndman et al. (2011) and Wickramasuriya et al. (2019). Although originally developed in the context of hierarchical time series, the methodology has since been extended to temporal (Athanasopoulos et al., 2017) and cross-temporal (Kourentzes & Athanasopoulos, 2019) settings. A geometric interpretation of the reconciliation framework is provided in Panagiotelis et al. (2021). A comprehensive and up-to-date review of the developments and results in this area is (Athanasopoulos et al., 2024).

The objective of this note is to provide a novel derivation of the forecast reconciliation formulae that clarifies the underlying assumptions, offers a more intuitive proof, based on optimal linear projection arguments, complements the geometric interpretation, and facilitates generalizations to partial reconciliation, which deals with situations where only a subset of the component series is reconciled. In the next section, we formulate the reconciliation problem and introduce the notation. The main result is presented in Section 3, while Section 4 extends it to partial reconciliation problems.

2. Problem statement and notation

2.1. The measurement model

Let $\mathbf{y}_t$ denote an n-dimensional time series, observed for $t = 1, 2, \ldots, T$. The first $k < n$ components, denoted $\mathbf{y}_{At}$, are obtained by aggregating the bottom $m = n - k$ components, denoted $\mathbf{b}_t$, which represent the disaggregate series. Hence, $\mathbf{y}_t$ is partitioned as $\mathbf{y}_t = [\mathbf{y}'_{At}, \mathbf{b}'_t]'$, where, given the $m \times k$ aggregation matrix $\mathbf{A}$, we write $\mathbf{y}_{At} = \mathbf{A}'\mathbf{b}_t$.

The above measurement framework occurs in the following situations.

- In the reconciliation of hierarchical time series, the matrix $\mathbf{A}$ is a known cross-sectional aggregation matrix applied to the m bottom-level time series $\mathbf{b}_t$. Fig.

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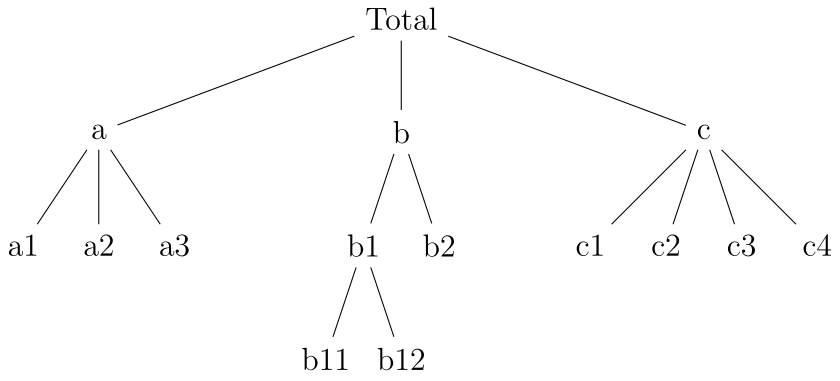


Fig. 1. Example of a hierarchical time series with $n = 15$ elements and $m = 10$ disaggregate series.

1 provides an instance of a hierarchical time series with $n = 15$ components. The $m = 10$ disaggregate series are

$$\mathbf{b}_t = (y_{a1,t}, y_{a2,t}, y_{a3,t}, y_{b11,t}, y_{b12,t}, y_{b2,t}, y_{c1,t}, y_{c2,t}, y_{c3,t}, y_{c4,t})',$$

whereas $\mathbf{y}_{At} = (y_{Total,t}, y_{a,t}, y_{b,t}, y_{b1,t}, y_{c,t})'$, has elements obtained from the sum of the of the child nodes, $\mathbf{y}_{At} = \mathbf{A}'\mathbf{b}_t$, with

$$\mathbf{A}' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- The matrix $\mathbf{A}$ is specified so that its columns are the first k eigenvectors of the variance–covariance matrix of the series $\mathbf{b}_t$. The vector $\mathbf{y}_{At}$ contains the first k principal components scores at time t of the series $\mathbf{b}_t$, see Yang et al. (2024). Alternatively, or in conjunction, $\mathbf{y}_{At}$ are k random linear projections, and $\mathbf{A}$ contains the coefficients of the projection.
- The matrix $\mathbf{A}$ is a linear temporal aggregation matrix (Athanasopoulos et al., 2017), or a cross-temporal aggregation matrix (Kourentzes & Athanasopoulos, 2019).

For general discussion of the above specifications, see Di Fonzo and Girolimetto (2024) and Athanasopoulos et al. (2024).

Introducing the matrices

$$\mathbf{S} = \begin{bmatrix} \mathbf{A}' \\ \mathbf{I}_m \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} \mathbf{0}' \\ \mathbf{I}_m \end{bmatrix},$$

where $\mathbf{0}$ is an $m \times k$ matrix of zeroes and $\mathbf{I}_m$ is the identity matrix of order m, the measurement model is represented as follows:

$$\mathbf{y}_t = \mathbf{S}\mathbf{b}_t, \tag{1}$$

with $\mathbf{b}_t = \mathbf{J}'\mathbf{y}_t$. The matrix $\mathbf{J}$ selects the bottom elements of $\mathbf{y}_t$. As can be seen by premultiplying both sides of (1) by $\mathbf{J}'$, it is related to $\mathbf{S}$ via the coherence constraints $\mathbf{J}'\mathbf{S} = \mathbf{I}_m$.

Adopting the terminology of Panagiotelis et al. (2021), the columns of the matrix $\mathbf{S}$ span the m-dimensional *coherent* subspace of $\mathbb{R}^n$, which will be denoted by $\mathcal{S}$.

As it is clear from (1), $\mathbf{y}_t \in \mathcal{S}$, and thus has a singular distribution. Letting $\mathbf{C}$ denote the $n \times k$ orthogonal complement of $\mathbf{S}$, $\mathbf{C}'\mathbf{S} = \mathbf{0}'$, it holds that $\mathbf{C}'\mathbf{y}_t = \mathbf{0}$, a vector of k zeros. The columns of $\mathbf{C}$ span the k-dimensional *constraint violation* subspace, denoted $\mathcal{S}_\perp$. This subspace is identifiable only up to a nonsingular $k \times k$ transformation, as will be shown next.

If $\mathbf{C}$ is partitioned as $\mathbf{C} = \begin{bmatrix} \mathbf{C}_A \\ \mathbf{C}_B \end{bmatrix}$, where $\mathbf{C}_A$ is $k \times k$ and $\mathbf{C}_B$ is $m \times k$, then the orthogonality condition $\mathbf{C}'\mathbf{S} = \mathbf{0}'$ imposes the mk restrictions $\mathbf{C}_B = -\mathbf{A}\mathbf{C}_A$. Consequently, $\mathbf{C}$ can be written as $\mathbf{C} = \begin{bmatrix} \mathbf{I}_k \\ -\mathbf{A} \end{bmatrix} \mathbf{C}_A$. However, the specific choice of $\mathbf{C}_A$ does not affect the reconciliation of the forecasts. Without loss of generality, in the sequel we set $\mathbf{C}_A = -\mathbf{I}_k$.

In the more general setting considered in Girolimetto and Di Fonzo (2024), $\mathbf{y}$ is a linearly constrained vector, partitioned so that the first k elements are identified as the constrained variables. At the same time, $\mathbf{b}_t$ is a set of free variables. The n series are subject to k non-redundant constraints, $\mathbf{\Gamma}'\mathbf{y}$, where $\mathbf{\Gamma} = [\mathbf{\Gamma}_A', \mathbf{\Gamma}_b']'$ is an $n \times k$ matrix with rank k and $\mathbf{\Gamma}_A$ is invertible. Then, representation (1) holds with $\mathbf{A} = -\mathbf{\Gamma}_b\mathbf{\Gamma}_A^{-1}$ in $\mathbf{S} = [\mathbf{A}, \mathbf{I}_m]'$.

2.2. Forecast reconciliation

Our objective is to predict $\mathbf{y}_{T+h}$ at time T, where h is the forecast horizon. Suppose we have an unbiased preliminary predictor, $\hat{\mathbf{y}}_{T+h|t}$, that lies in an n-dimensional space. To improve forecast accuracy, we aim to reconcile this predictor by enforcing the aggregation constraints. Specifically, we seek a reconciled forecast, $\tilde{\mathbf{y}}_{T+h|t}$, that satisfies the coherence condition $\mathbf{C}'\tilde{\mathbf{y}}_{T+h|t} = \mathbf{0}$.

The mechanism of projecting the preliminary forecast onto the coherent subspace is discussed in the next section. We will show that the reconciled forecast is obtained by subtracting from $\hat{\mathbf{y}}_{T+h|t}$ the component of the prediction error that lies in the constraint violation subspace. As a result, $\tilde{\mathbf{y}}_{T+h|t}$ is more accurate than the original forecast $\hat{\mathbf{y}}_{T+h|t}$, in terms of mean squared error.

The reconciliation formulae relating $\tilde{\mathbf{y}}_{T+h|t}$ to $\hat{\mathbf{y}}_{T+h|t}$ were derived by Hyndman et al. (2011) Wickramasuriya et al. (2019). A comprehensive overview of key applications is provided by Athanasopoulos et al. (2024). The topic is also closely related to the literature on time se-

ries benchmarking, for which Dagum and Cholette (2006) offer a thorough review.

Against this background, the next section provides an alternative derivation of the optimal reconciled predictor, based on optimal linear projection arguments, such that the reconciliation error is distilled from the preliminary prediction error. The proof is based on minimal assumptions concerning the role of the reconciliation error and its relationship with the preliminary prediction error, thereby clarifying some aspects of the theory of forecast reconciliation.

3. Main result

For notational simplicity, in the sequel, we will drop the subscripts referring to time and forecast horizon. Hence, we denote $\mathbf{y} = \mathbf{y}_{T+h}$, $\hat{\mathbf{y}} = \hat{\mathbf{y}}_{T+h|t}$, and $\tilde{\mathbf{y}} = \tilde{\mathbf{y}}_{T+h|t}$. A similar convention is adopted for $\mathbf{b} = \mathbf{b}_{T+h}$.

The preliminary predictor, $\hat{\mathbf{y}}$, originates from a model or a method that does not take into account the aggregation constraints binding the series. This occurs, for instance, when the individual time series are forecasted separately. The prediction error has a non-singular distribution, since a component of the prediction error lies in the nullity of the coherent subspace.

Assumption 1. The preliminary predictor $\hat{\mathbf{y}}$ is such that $\mathbf{E}(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$ and $\mathbf{E}\{(\mathbf{y} - \hat{\mathbf{y}})(\mathbf{y} - \hat{\mathbf{y}})'\} = \boldsymbol{\Omega}$ has full rank.

In the sequel, we denote the preliminary prediction error by $\hat{\mathbf{v}} = \mathbf{y} - \hat{\mathbf{y}}$. The unbiasedness of $\hat{\mathbf{y}}$ is convenient, but relaxing it does not affect the effectiveness of reconciliation in reducing the mean square forecast error. The rank condition on $\boldsymbol{\Omega}$ will be used in the derivation of the main result. However, it is stronger than strictly necessary and, in the case of partially reconciled forecasts (dealt with by the next section), it can be relaxed to $\text{rank}(\boldsymbol{\Omega}) \geq \max\{m, k\}$, so that $\mathbf{C}'\boldsymbol{\Omega}\mathbf{C}$ or $\mathbf{S}'\boldsymbol{\Omega}^{-}\mathbf{S}$ are nonsingular, where $\boldsymbol{\Omega}^{-}$ is a generalized (Moore–Penrose) inverse of $\boldsymbol{\Omega}$.

Let $\tilde{\mathbf{y}}$ denote the reconciled predictor, which satisfies the coherence conditions $\tilde{\mathbf{y}} = \mathbf{S}\mathbf{J}'\tilde{\mathbf{y}}$ and $\mathbf{C}'\tilde{\mathbf{y}} = \mathbf{0}$. The corresponding reconciled prediction error is defined as $\tilde{\mathbf{v}} = \mathbf{y} - \tilde{\mathbf{y}}$.

Assumption 2. The reconciled predictor is related to the preliminary one by

$$\hat{\mathbf{y}} = \tilde{\mathbf{y}} + \boldsymbol{\epsilon}. \tag{2}$$

The reconciliation error, $\boldsymbol{\epsilon}$, has zero mean and is uncorrelated with $\tilde{\mathbf{v}}$. Its distribution is singular with $\text{Var}(\boldsymbol{\epsilon}|\mathbf{C}'\boldsymbol{\epsilon}) = \mathbf{0}$.

According to Assumption 2, the preliminary and reconciled predictors differ only due to the reconciliation error. The latter lies in a subspace of the preliminary prediction error $\hat{\mathbf{v}}$. The singularity of $\boldsymbol{\epsilon}$ formalizes the notion that the reconciliation error lies in the space orthogonal to the coherent subspace (so that its projection onto it is the zero vector). The condition $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{C}'\boldsymbol{\epsilon}) = \mathbf{0}$ implies that there exists an $n \times k$ matrix $\boldsymbol{\Theta}$ such that $\boldsymbol{\epsilon} = \boldsymbol{\Theta}\mathbf{C}'\boldsymbol{\epsilon}$ and $\mathbf{C}'\boldsymbol{\Theta} = \mathbf{I}_k$.

Subtracting $\mathbf{y}$ from both sides of Eq. (2) and rearranging, yields $\boldsymbol{\epsilon} = \tilde{\mathbf{v}} - \hat{\mathbf{v}}$. Also, premultiplying both sides by $\mathbf{C}'$, $\mathbf{C}'\hat{\mathbf{v}} = -\mathbf{C}'\boldsymbol{\epsilon}$, and $\mathbf{C}'\hat{\mathbf{y}} = \mathbf{C}'\boldsymbol{\epsilon}$. In other words, $\boldsymbol{\epsilon}$ is a component of the preliminary prediction error, lying in the constraint violation subspace, that has to be extracted and subtracted from $\hat{\mathbf{y}}$. On the contrary, $\tilde{\mathbf{v}}$ lies in the coherent subspace.

Theorem 1. *The best linear reconciled predictor, $\tilde{\mathbf{y}}$, is the linear transformation of the preliminary predictor,*

$$\tilde{\mathbf{y}} = \mathbf{M}\hat{\mathbf{y}}, \quad \mathbf{M} = \mathbf{I} - \boldsymbol{\Omega}\mathbf{C}(\mathbf{C}'\boldsymbol{\Omega}\mathbf{C})^{-1}\mathbf{C}'. \tag{3}$$

Also, denoting $\mathbf{V} = E\{(\mathbf{y} - \tilde{\mathbf{y}})(\mathbf{y} - \tilde{\mathbf{y}})'\}$, it holds

$$\mathbf{V} = \boldsymbol{\Omega} - \boldsymbol{\Omega}\mathbf{C}(\mathbf{C}'\boldsymbol{\Omega}\mathbf{C})^{-1}\mathbf{C}'\boldsymbol{\Omega}. \tag{4}$$

Equivalent expressions for $\mathbf{M}$ and $\mathbf{V}$ are

$$\mathbf{M} = \mathbf{S}(\mathbf{S}'\boldsymbol{\Omega}^{-1}\mathbf{S})^{-1}\mathbf{S}'\boldsymbol{\Omega}^{-1}, \quad \mathbf{V} = \mathbf{S}(\mathbf{S}'\boldsymbol{\Omega}^{-1}\mathbf{S})^{-1}\mathbf{S}'. \tag{5}$$

Proof. The discrepancy $\mathbf{e} = \mathbf{C}'\boldsymbol{\epsilon}$ has mean zero and a variance–covariance matrix $\mathbf{C}'\boldsymbol{\Omega}\mathbf{C}$. Moreover,

$$\begin{aligned}\text{Cov}(\boldsymbol{\epsilon}, \mathbf{e}) &= \mathrm{E}\{(\tilde{\mathbf{v}} - \hat{\mathbf{v}})\boldsymbol{\epsilon}'\mathbf{C}\} \\ &= \boldsymbol{\Omega}\mathbf{C},\end{aligned}$$

since $\boldsymbol{\epsilon}$ is orthogonal to $\tilde{\mathbf{v}}$, and $\mathbf{e} = -\mathbf{C}'\hat{\mathbf{v}}$. The best linear projection of $\boldsymbol{\epsilon}$ on $\mathbf{e}$ is

$$\begin{aligned}\hat{\boldsymbol{\epsilon}} &= \text{Cov}(\boldsymbol{\epsilon}, \mathbf{e})\text{Var}^{-1}(\mathbf{e})\mathbf{e} \\ &= \boldsymbol{\Omega}\mathbf{C}(\mathbf{C}'\boldsymbol{\Omega}\mathbf{C})^{-1}\mathbf{C}'\hat{\mathbf{y}}.\end{aligned}$$

Subtracting $\hat{\boldsymbol{\epsilon}}$ from $\hat{\mathbf{y}}$, in the light of (2), gives the result (3). Noticing that $\mathbf{M}\mathbf{y} = \mathbf{0}$, $\tilde{\mathbf{v}} = \mathbf{M}\hat{\mathbf{v}}$, and thus $\mathbf{V} = \mathbf{M}\boldsymbol{\Omega}\mathbf{M}'$, which gives (4).

Finally, statement (5) is proved by noticing $\mathbf{M}\tilde{\mathbf{y}} = \tilde{\mathbf{y}}$ and that premultiplying both sides of $\tilde{\mathbf{y}} = \mathbf{M}\hat{\mathbf{y}}$ by $\mathbf{S}'\boldsymbol{\Omega}^{-1}\mathbf{M}$ gives $\mathbf{S}'\boldsymbol{\Omega}^{-1}\tilde{\mathbf{y}} = \mathbf{S}'\boldsymbol{\Omega}^{-1}\hat{\mathbf{y}}$ (these linear projections are free from reconciliation error). Replacing $\tilde{\mathbf{y}} = \mathbf{S}\mathbf{J}'\tilde{\mathbf{y}}$ and rearranging gives $\mathbf{J}'\tilde{\mathbf{y}} = (\mathbf{S}'\boldsymbol{\Omega}^{-1}\mathbf{S})^{-1}\mathbf{S}'\boldsymbol{\Omega}^{-1}\hat{\mathbf{y}}$. Premultiplying both sides by $\mathbf{S}$ gives the first expression in (5). The second expression follows from $\mathbf{V} = \mathbf{M}\boldsymbol{\Omega}\mathbf{M}'$, with $\mathbf{M}$ given in (5). □

Reconciliation improves forecast accuracy, as $\boldsymbol{\Omega} - \mathbf{V}$ equals a positive semi-definite matrix, see (4). A key driver of accuracy is $\boldsymbol{\Omega}$, which, however, needs to be estimated. For a review of the possibilities, see Athanasopoulos et al. (2024) and the references therein. The matrix $\mathbf{M}$ is the idempotent matrix, projecting $\hat{\mathbf{y}}$ into the coherent subspace along the constraint violation space. It is factored as $\mathbf{S}\mathbf{G}$, where $\mathbf{G} = (\mathbf{S}'\boldsymbol{\Omega}^{-1}\mathbf{S})^{-1}\mathbf{S}'\boldsymbol{\Omega}^{-1}$ is a generalized inverse of $\mathbf{S}$. The following corollary deals with the optimality of the solution (3)–(5).

Corollary 1. *The vector $\tilde{\mathbf{y}}_B = \mathbf{G}\hat{\mathbf{y}}$, is the unique minimizer of $\|\mathbf{S}\mathbf{y}_B - \hat{\mathbf{y}}\|^2_{\Omega^{-1}} = (\mathbf{S}\mathbf{y}_B - \hat{\mathbf{y}})'\boldsymbol{\Omega}^{-1}(\mathbf{S}\mathbf{y}_B - \hat{\mathbf{y}})$ with respect to $\mathbf{y}_B$.*

Proof. The proof follows directly from Theorem 3.3.1 in Rao and Mitra (1971), as $\mathbf{S}\mathbf{G}\mathbf{S} = \mathbf{S}$, $\mathbf{G}\mathbf{S}\mathbf{G} = \mathbf{G}$, $\mathbf{G}\mathbf{S} = \mathbf{I}_m$ and $\mathbf{G}'\mathbf{S}'\boldsymbol{\Omega}^{-1} = \boldsymbol{\Omega}^{-1}\mathbf{S}\mathbf{G}$, see conditions (i) in Note 3, page 52 in Rao and Mitra (1971). □

Note that (5) is equivalently obtained by generalized least square estimation in $\hat{\mathbf{y}} = \mathbf{S}\boldsymbol{\beta} - \hat{\mathbf{v}}$, where $\boldsymbol{\beta} = \mathbf{J}'\mathbf{y}$, which was the original approach in Hyndman et al. (2011). Wickramasuriya et al. (2019) derive $\mathbf{G}$ as the minimizer of the trace of $\mathbf{S}\mathbf{G}\boldsymbol{\Omega}\mathbf{G}\mathbf{S}'$.

Remark. The linear optimal reconciled predictor is based on the oblique projection of $\hat{\mathbf{y}}$ on the coherent space.

The reconciled predictor resulting from the orthogonal projection of the preliminary estimates into the coherent subspace, $\tilde{\mathbf{y}}_\perp = \mathbf{S}(\mathbf{S}'\mathbf{S})^{-1}\mathbf{S}'\hat{\mathbf{y}}$, equivalently $\tilde{\mathbf{y}}_\perp = [\mathbf{I} - \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}']\hat{\mathbf{y}}$, is optimal only in the case $\boldsymbol{\Omega} = \sigma^2\mathbf{I}$. For general $\boldsymbol{\Omega}$, $\tilde{\mathbf{y}}_\perp$ is admissible, in the sense that it leads to a reconciled estimate satisfying $\mathbf{C}'\tilde{\mathbf{y}}_\perp = \mathbf{0}$; still, it is suboptimal, as it relies on an alternative estimator of $\boldsymbol{\epsilon}$, $\hat{\boldsymbol{\epsilon}}_\perp = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}'\hat{\mathbf{y}}$, characterized by a larger estimation error variance.

The variance of the corresponding prediction error, $\mathbf{V}_\perp = \mathrm{Var}(\mathbf{y} - \tilde{\mathbf{y}}_\perp)$, differs from $\mathbf{V}$, given in (5), by a positive semi-definite matrix:

$$\begin{aligned}\mathbf{V}_\perp - \mathbf{V} &= \mathbf{S}(\mathbf{S}'\mathbf{S})^{-1}\mathbf{S}'\boldsymbol{\Omega}\mathbf{S}(\mathbf{S}'\mathbf{S})^{-1}\mathbf{S}' - \mathbf{S}(\mathbf{S}'\boldsymbol{\Omega}^{-1}\mathbf{S})^{-1}\mathbf{S}' \\ &= \mathbf{S}(\mathbf{S}'\mathbf{S})^{-1}[\mathbf{S}'\boldsymbol{\Omega}\mathbf{S} - \mathbf{S}'\mathbf{S}(\mathbf{S}'\boldsymbol{\Omega}^{-1}\mathbf{S})^{-1}\mathbf{S}'\mathbf{S}](\mathbf{S}'\mathbf{S})^{-1}\mathbf{S}' \\ &\geq \mathbf{0}\end{aligned}$$

The last line follows from setting $\mathbf{E} = \boldsymbol{\Omega}^{1/2}\mathbf{S}$, $\mathbf{F} = \boldsymbol{\Omega}^{-1/2}\mathbf{S}$, so that $\mathbf{E}'\mathbf{E} = \mathbf{S}'\boldsymbol{\Omega}\mathbf{S}$, $\mathbf{F}'\mathbf{F} = \mathbf{S}'\boldsymbol{\Omega}^{-1}\mathbf{S}$, $\mathbf{E}'\mathbf{F} = \mathbf{S}'\mathbf{S}$, and applying the matrix Cauchy–Schwarz inequality (Pečarić et al., 1996), $\mathbf{E}'\mathbf{E} \geq \mathbf{E}'\mathbf{F}(\mathbf{F}'\mathbf{F})^{-1}\mathbf{F}'\mathbf{E}$. This shows that the matrix in square brackets is positive semi-definite, and thus $\mathbf{V}_\perp \geq \mathbf{V}$. △

Finally, since $\mathrm{Var}(\boldsymbol{\epsilon}) = \mathrm{E}\{\mathrm{Var}(\boldsymbol{\epsilon}|\mathbf{C}'\boldsymbol{\epsilon})\} + \mathrm{Var}\{\mathrm{E}(\boldsymbol{\epsilon}|\mathbf{C}'\boldsymbol{\epsilon})\}$, the following corollary holds.

Corollary 2. *Assuming further that $\boldsymbol{\epsilon}$ has a Gaussian distribution, the reconciliation error has mean zero and covariance matrix $Var(\boldsymbol{\epsilon}) = \boldsymbol{\Omega}\mathbf{C}(\mathbf{C}'\boldsymbol{\Omega}\mathbf{C})^{-1}\mathbf{C}'\boldsymbol{\Omega}$.*

Proof. Under the Gaussian assumption, $\hat{\boldsymbol{\epsilon}} = \mathrm{E}(\boldsymbol{\epsilon}|\mathbf{C}'\boldsymbol{\epsilon})$ (the optimal linear estimator coincides with the conditional expectation). Hence, $\mathrm{Var}\{\mathrm{E}(\boldsymbol{\epsilon}|\mathbf{C}'\boldsymbol{\epsilon})\} = \boldsymbol{\Omega}\mathbf{C}(\mathbf{C}'\boldsymbol{\Omega}\mathbf{C})^{-1}\mathbf{C}'\boldsymbol{\Omega}$ and by Assumption 2, $\mathrm{Var}(\boldsymbol{\epsilon}|\mathbf{C}'\boldsymbol{\epsilon}) = \mathbf{0}$. □

4. Bottom-up, top-down predictors and best linear disaggregation

The results in the previous section encompass methods of partial reconciliation, which deal with situations in which only a subset of the components of $\mathbf{y}$ are subject to reconciliation. These include the single-level approaches reviewed in Athanasopoulos et al. (2024, Section 2.4) and the linear disaggregation problem discussed in Section 4.3. In the Appendix, we review the related problem of forecast reconciliation with *immutable forecasts*, considered by Zhang et al. (2023), which provides a specific instance in which the forecasts of a designated subset of time series are constrained to remain unchanged.

4.1. Bottom-up predictor

The bottom-up predictor, denoted $\tilde{\mathbf{y}}^*$, is obtained by predicting $\mathbf{y}_A$ through aggregation of the disaggregate forecasts $\hat{\mathbf{b}} = \mathbf{J}'\hat{\mathbf{y}}$, i.e., $\tilde{\mathbf{y}}^*_A = \mathbf{A}'\hat{\mathbf{b}}$. This corresponds to setting $\tilde{\mathbf{y}}^* = \mathbf{S}\mathbf{J}'\hat{\mathbf{y}}$, which satisfies the coherence constraints, $\mathbf{C}'\tilde{\mathbf{y}}^* = \mathbf{0}$.

Adding and subtracting $\mathbf{y} = \mathbf{S}\mathbf{J}'\mathbf{y}$, the prediction error of the bottom-up predictor, $\tilde{\mathbf{v}}^* = \mathbf{y} - \tilde{\mathbf{y}}^*$, can be written as $\tilde{\mathbf{v}}^* = \mathbf{S}\mathbf{J}'\hat{\mathbf{v}}$. Consequently, the variance of $\tilde{\mathbf{v}}^*$ is $\mathbf{V}^* = \mathbf{S}\mathbf{J}'\boldsymbol{\Omega}\mathbf{J}\mathbf{S}'$, where $\mathbf{J}'\boldsymbol{\Omega}\mathbf{J} = \boldsymbol{\Omega}_B$ is the mean square error matrix of $\hat{\mathbf{b}}$.

Now, partition the reconciliation residual $\boldsymbol{\epsilon} = \hat{\mathbf{y}} - \tilde{\mathbf{y}}^*$ as $\boldsymbol{\epsilon} = [\boldsymbol{\epsilon}'_A, \boldsymbol{\epsilon}'_B]'$. Since $\tilde{\mathbf{y}}^* = \mathbf{S}\mathbf{J}'\hat{\mathbf{y}}$, it follows that $\boldsymbol{\epsilon} = (\mathbf{I} - \mathbf{S}\mathbf{J}')\hat{\mathbf{y}}$, which yields $\boldsymbol{\epsilon}_B = \mathbf{0}$ and $\boldsymbol{\epsilon}_A = -\mathbf{C}'\hat{\mathbf{y}}$. Interestingly, the bottom up approach can be recovered by replacing $\boldsymbol{\Omega}^{-1}$ with

$$\boldsymbol{\Omega}^- = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Omega}_B^{-1} \end{bmatrix}$$

in Eq. (5).

4.2. Top-down predictor

The classical top-down approach, as documented in Gross and Sohl (1990), (Dangerfield & Morris, 1992), Athanasopoulos et al. (2009), Hyndman et al. (2011), and Panagiotelis et al. (2021), among others, applies when the univariate aggregate top-level forecast is taken as given ($\tilde{y}_A = \hat{y}_A$), and disaggregated without further adjustment. When the top-level vector $\mathbf{y}_A$ contains multiple aggregates subject to internal constraints, we assume that $\tilde{\mathbf{y}}_A = \hat{\mathbf{y}}_A$ holds after any prior reconciliation applied to the aggregate series. The difference with the reconciliation problem with immutable forecasts is discussed in Appendix.

The top-down predictor entails disaggregating $\hat{\mathbf{y}}_A$ to obtain predictions for the disaggregate series $\mathbf{b}$. The motivation for adopting a top-down approach is often twofold: either the disaggregate series $\mathbf{b}$ are not observed, or forecasting the high-frequency series directly is impractical or less reliable.

Let us distinguish two cases. In the first, assume that the aggregate series is related to the disaggregate series via $\mathbf{y}_A = \mathbf{Q}\mathbf{b}$, where $\mathbf{Q}$ is a $k \times m$ matrix, $k > m$, with rank not greater than m. Thus, $\mathbf{y}_A$ lies in a subset of the m-dimensional subspace. This setting arises, for example, in temporal hierarchies, such as when annual, semiannual, four-monthly, quarterly, and bimonthly aggregates are constructed in a way that allows recovery of $\mathbf{b}$ from $\mathbf{y}_A$. The reconciled forecast of $\mathbf{b}$ is given by $\tilde{\mathbf{b}} = \mathbf{Q}^-\tilde{\mathbf{y}}_A$, where $\mathbf{Q}^-$ denotes a pseudo-inverse of $\mathbf{Q}$.

Example. Consider a grouped structure with two cross-sectional units, represented by the regions North (N) and South (S), each of which manufactures two products, labeled 1 and 2. Let y_T denote the aggregate output, resulting both from the sum of regional outputs, $y_T = y_N + y_S$, and from the sum of total output of the two products, $y_T = y_1 + y_2$. The aggregate series are stacked in the 5×1 vector $\mathbf{y}_A = [y_T, y_N, y_S, y_1, y_2]'$, and there are four bottom series, $\mathbf{b} = [y_{N1}, y_{N2}, y_{S1}, y_{S2}]'$, where y_{Ri} is the output of product $i = \{1, 2\}$ in region $R = \{N, S\}$, subject to the aggregation constraints $y_N = y_{N1} + y_{N2}$, $y_S = y_{S1} + y_{S2}$, and $y_1 = y_{N1} + y_{S1}$, $y_2 = y_{N2} + y_{S2}$.

The aggregate series are related to the bottom series by $\mathbf{y}_A = \mathbf{Q}\mathbf{b}$, where $\mathbf{Q}$ is the rank 3 matrix

$$\mathbf{Q} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Define the matrix $\mathbf{C}_A$, such that $\mathbf{C}'_A\mathbf{y}_A = \mathbf{0}$,

$$\mathbf{C}'_A = \begin{bmatrix} 1 & -1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \end{bmatrix},$$

(note also that $\mathbf{C}_A'\mathbf{Q} = \mathbf{0}$) and denote by $\hat{\mathbf{y}}_A$ the preliminary prediction of the top level series, with $\Omega_A = \mathrm{Var}(\mathbf{y}_A - \hat{\mathbf{y}}_A)$. The reconciled top-level predictions are

$$\tilde{\mathbf{y}}_A = \hat{\mathbf{y}}_A - \Omega_A\mathbf{C}_A(\mathbf{C}_A'\Omega_A\mathbf{C}_A)^{-1}\mathbf{C}_A'\hat{\mathbf{y}}_A.$$

The bottom-level predictions are then obtained as $\tilde{\mathbf{b}} = \mathbf{Q}^-\tilde{\mathbf{y}}_A$ where $\mathbf{Q}^-$ is a pseudo-inverse of $\mathbf{Q}$. △

If $\mathbf{Q}'\Omega_A\mathbf{Q}$ is nonsingular, with $\Omega_A = \mathrm{Var}(\mathbf{y}_A - \hat{\mathbf{y}}_A)$, a suitable choice for the pseudo inverse is $\mathbf{Q}^- = (\mathbf{Q}'\Omega_A^{-1}\mathbf{Q})^{-1}\mathbf{Q}'\Omega_A^{-1}$. This case fits the general framework introduced earlier, with:

$$\mathbf{J} = \begin{bmatrix} \mathbf{I}_k \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} \mathbf{I}_k \\ \mathbf{Q}^- \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \mathbf{Q}^{-'} \\ -\mathbf{I}_m \end{bmatrix}.$$

The top-down solution is also recovered by applying Eq. (5) with:

$$\Omega^- = \begin{bmatrix} \Omega_A^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix},$$

or, equivalently, $\Omega^- = \mathbf{J}\Omega_A^{-1}\mathbf{J}'$. This case is structurally symmetric to the bottom-up approach, with the roles of $\mathbf{y}_A$ and $\mathbf{b}$ reversed. As in the bottom-up case, the projection matrix is $\mathbf{M} = \mathbf{S}\mathbf{J}'$.

The case when $k < m$, or more generally when $\mathbf{y}_A$ lies in a k^*-dimensional subspace of $\mathbb{R}^m$, $k^* \leq k$, is more problematic. In this setting, distributing the prediction $\hat{\mathbf{y}}_A$ across the components of $\mathbf{b}$ inherently involves an element of arbitrariness. This ambiguity can be addressed by placing a prior distribution on $\mathbf{b}$, such that $\mathbf{b} \sim (\boldsymbol{\mu}_B, \boldsymbol{\Sigma}_B)$, where $\boldsymbol{\mu}_B$ and $\boldsymbol{\Sigma}_B$ denote the mean and covariance matrix of $\mathbf{b}$. Under this prior, the optimal linear prediction of $\mathbf{b}$ given $\hat{\mathbf{y}}_A$ is

$$\tilde{\mathbf{b}} = [\mathbf{I}_m - \boldsymbol{\Sigma}_B\mathbf{A}(\mathbf{A}'\boldsymbol{\Sigma}_B\mathbf{A})^{-1}\mathbf{A}']\boldsymbol{\mu}_B + \boldsymbol{\Sigma}_B\mathbf{A}(\mathbf{A}'\boldsymbol{\Sigma}_B\mathbf{A})^{-1}\hat{\mathbf{y}}_A.$$

Note that $\mathbf{A}'\tilde{\mathbf{b}} = \hat{\mathbf{y}}_A$, and thus the prediction of $\mathbf{b}$ satisfies the linear constraints; the final top-down estimate, $\tilde{\mathbf{y}} = [\hat{\mathbf{y}}_A', \tilde{\mathbf{b}}']'$, is also be obtained from (3), by setting

$$\hat{\mathbf{y}} = \begin{bmatrix} \hat{\mathbf{y}}_A \\ \boldsymbol{\mu}_B \end{bmatrix}, \quad \Omega = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_B \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{I}_k \\ \mathbf{A} \end{bmatrix}.$$

A special case can be considered, where $\boldsymbol{\mu}_B = \mathbf{0}$ and $\boldsymbol{\Sigma}_B = \mathrm{diag}(w_1, \ldots, w_m)$, with w_j representing a measure of importance or weight assigned to the jth component of $\mathbf{b}$. These weights may reflect size, contribution to the aggregate, or frequency of occurrence. When the aggregation corresponds to a total, i.e., $\mathbf{A} = [1, 1, \ldots, 1]'$, so that $y_A = \sum_j b_j$, the aggregate forecast $\hat{y}_A$ is distributed to the lowest level series according to the proportion $p_j = w_j/\sum_k w_k$, giving $\tilde{b}_j = p_j\hat{y}_A, \quad j = 1, \ldots, m$. Notice that if $\boldsymbol{\mu}_B = \mathbf{0}$, then, for any arbitrary choice of $\boldsymbol{\Sigma}_B$, $\tilde{\mathbf{b}} = \mathbf{P}\hat{\mathbf{y}}_A$, $\mathbf{P} = \boldsymbol{\Sigma}_B\mathbf{A}(\mathbf{A}'\boldsymbol{\Sigma}_B\mathbf{A})^{-1}$, lies in a k-dimensional subspace and $\mathbf{P}$ is not a projection matrix, but only the right inverse of $\mathbf{A}'$. In this case, the reconciled forecast violates the unbiasedness property defined by Hyndman et al. (2011) and Panagiotelis et al. (2021, Theorem 4.1).

4.3. Linear disaggregation

The linear disaggregation problem concerns a partially observed vector $\mathbf{y}$, where the aggregate component $\mathbf{y}_A$ is known, while only a preliminary predictor $\hat{\mathbf{b}}$ is available for the disaggregate component $\mathbf{b}$. Consider, for instance, the case in which $\mathbf{y}_A$ is a $k \times 1$ vector of annual value added, observed for k consecutive years, and $\mathbf{b}$ is a $m = 4k$ vector of quarterly disaggregated value added, such that the sum of the four quarters equals the annual total. The aggregation matrix is $\mathbf{A} = \mathbf{i}_4 \otimes \mathbf{I}_k$, where $\mathbf{i}_4 = [1\ 1\ 1\ 1]'$. The reconciliation objective is to improve the preliminary predictor $\hat{\mathbf{b}}$ by incorporating the aggregate information $\mathbf{y}_A = \mathbf{A}'\mathbf{b}$.

Regarding $\hat{\mathbf{b}}$, we assume that it is given exogenously, as in Denton (1971). In contrast, in Chow and Lin (1971) $\hat{\mathbf{b}}$ is constructed as a linear combination of related series, with the combination coefficients estimated via a generalized regression of the aggregate series on the aggregation of the related indicators. The goal is to obtain an improved predictor of the disaggregated series, denoted $\tilde{\mathbf{b}}^+$, that incorporates information from the observed aggregate $\mathbf{y}_A$.

In this setting, the preliminary forecast is represented by the vector $\hat{\mathbf{y}} = [\mathbf{y}_A', \hat{\mathbf{b}}']'$, while the reconciled (or disaggregated) forecast takes the form $\tilde{\mathbf{y}} = [\mathbf{y}_A', \tilde{\mathbf{b}}^{+'}]'$. As a result, $\hat{\mathbf{v}}_A = \mathbf{0}$ and $\epsilon_A = 0$, while aggregation of the disaggregate forecast $\hat{\mathbf{b}}$ yields the discrepancy $\mathbf{e} = \mathbf{y}_A - \mathbf{A}'\hat{\mathbf{b}}$. Accordingly, it follows that $\mathbf{A}'\boldsymbol{\epsilon}_B = -\mathbf{A}'\hat{\mathbf{v}}_B = -\mathbf{e}$.

Given that $\mathrm{Var}(\mathbf{e}) = \mathbf{A}'\Omega_B\mathbf{A}$ and $\mathrm{Cov}(\boldsymbol{\epsilon}_B, \mathbf{e}) = \Omega_B\mathbf{A}$, the best linear predictor of $\mathbf{b}$ is obtained by redistributing the discrepancy across the disaggregate components:

$$\tilde{\mathbf{b}}^+ = \hat{\mathbf{b}} + \Omega_B\mathbf{A}(\mathbf{A}'\Omega_B\mathbf{A})^{-1}(\mathbf{y}_A - \mathbf{A}'\hat{\mathbf{b}}). \tag{6}$$

This is the classical optimal temporal disaggregation formula derived in Denton (1971) and Chow and Lin (1971), though obtained using different approaches. In the former reference, $\hat{\mathbf{b}}$ is given exogenously, and Ω_B is the inverse of a penalty matrix specified a priori by the investigator, which embodies a smoothness prior on the disaggregated estimates. In the latter, Ω_B is also specified a priori (e.g., as the autocovariance matrix of an autoregressive process). The formula is also derived in Byron (1978, 1979) in the context of balancing national accounts estimates.

The solution (6) corresponds to a special case of Eq. (3), obtained by specifying

$$\Omega = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \Omega_B \end{bmatrix}.$$

The mean squared error matrix of $\tilde{\mathbf{b}}^+$ can likewise be derived from Eq. (4).

5. Conclusions

Forecast reconciliation is an effective way of enforcing predictive accuracy in practical situations. This note has offered a new proof and additional insight into the nature and properties of the reconciliation problem.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT to check the English form of some sentences. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the publication's content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Forecast reconciliation with immutable forecasts

Reconciliation with immutable forecasts can be dealt with as a special case of Theorem 1. Such treatment has been provided by Zhang et al. (2023), and is adapted here to the notation used in this paper.

Consider the $n_z \times 1$ vector $\mathbf{z} = [\mathbf{y}_A', \mathbf{b}_z']'$, where $\mathbf{b}_z$ is a vector containing m_z basis time series (possibly from multiple levels of a hierarchical time series), partitioned as $\mathbf{b}_z = [\mathbf{b}', \mathbf{u}']'$, where $\mathbf{u}$ is an $r \times 1$ vector ($r < m_z$), whose forecasts are immutable, i.e. they are constrained to be unchanged under reconciliation, so that $\tilde{\mathbf{u}} = \hat{\mathbf{u}}$, while the remaining components (basis and non-basis) will adjust; $\mathbf{b}$ has $m = m_z - r$ elements.

Furthermore, it is assumed that $k \times 1$ vector $\mathbf{y}_A$ is a linear tranformation of the basis series, $\mathbf{y}_A = \mathbf{A}'\mathbf{b}_z$, so that, after a suitable partitioning of the matrix $\mathbf{A}$, $\mathbf{y}_A = \mathbf{A}_1'\mathbf{b} + \mathbf{A}_2'\mathbf{u}$.

Since the analysis is conditional on $\mathbf{u}$, Zhang et al. (2023) consider the subsystem consisting of $n = n_z - r$ time series

$$\begin{aligned}\mathbf{y} &= \begin{bmatrix} \mathbf{I}_k, & \mathbf{0}, & -\mathbf{A}_2' \\ \mathbf{0}, & \mathbf{I}_m, & \mathbf{0} \end{bmatrix} \mathbf{z} \\ &= \begin{bmatrix} \mathbf{y}_A - \mathbf{A}_2'\mathbf{u} \\ \mathbf{b} \end{bmatrix},\end{aligned}$$

admitting the representation $\mathbf{y} = \mathbf{S}\mathbf{b}$, with $\mathbf{b} = \mathbf{J}'\mathbf{y}$ and $\mathbf{C}'\mathbf{y} = \mathbf{0}$, where

$$\mathbf{S} = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{I}_m \end{bmatrix}, \mathbf{J} = \begin{bmatrix} \mathbf{0} \\ \mathbf{I}_m \end{bmatrix}, \mathbf{C} = \begin{bmatrix} \mathbf{I} \\ -\mathbf{A}_1 \end{bmatrix}.$$

This fits the general measurement framework of Section 2. Letting $\mathbf{\Omega} = \mathrm{Var}(\mathbf{y} - \hat{\mathbf{y}}|\mathbf{u} = \hat{\mathbf{u}})$, formulae (3)–(5) in Theorem 1 provide the way of transforming the prediction $\hat{\mathbf{y}} = [(\hat{\mathbf{y}}_A - \mathbf{A}_2'\hat{\mathbf{u}})', \hat{\mathbf{b}}']'$ into reconciled forecasts $\tilde{\mathbf{y}}$.

Thence, recalling $\tilde{\mathbf{u}} = \hat{\mathbf{u}}$, the full reconciled solution is

$$\tilde{\mathbf{z}} = \begin{bmatrix} \mathbf{I}_k, & \mathbf{0}, & \mathbf{A}_2' \\ \mathbf{0}, & \mathbf{I}_m, & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_r \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{y}} \\ \tilde{\mathbf{u}} \end{bmatrix}.$$

The immutable forecast framework is an instance of partial reconciliation, but the top-down approach cannot be considered as a particular case. Suppose that $\mathbf{A}_1 = \mathbf{0}$. It follows that $\mathbf{y}_A = \mathbf{A}_2\mathbf{u}$, and, while $\tilde{\mathbf{y}}_A = \hat{\mathbf{y}}_A$, $\mathbf{y}_A$ and $\mathbf{b}$ become decoupled, and the top block of $\mathbf{y}$ is a vector of zeroes. Section 4.2 deals with the case when $\mathbf{y}_A = \mathbf{A}'\mathbf{b}$ and $\tilde{\mathbf{y}}_A = \hat{\mathbf{y}}_A$.

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